FORCE™ Platform Delivers Exon Skipping PMO, Leads to Durable Increases in Dystrophin Protein in *mdx* Mice and Is Well Tolerated in NHPs

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Muscle Study Group Annual Scientific Meeting | Oct. 1, 2021

Ravi, *living with DMD*

Forward-Looking Statements

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This presentation also contains estimates, projections and other statistical data made by independent parties and by the Company relating to market size and growth and other data about the Company's industry and business. This data involves a number of assumptions and limitations, and you are cautioned not to give undue weight to such estimates. The Company has not independently verified the accuracy and completeness of the information obtained by third parties included in this presentation. In addition, projections, assumptions and estimates of the Company's future performance and the future performance of the markets in which the Company operates are necessarily subject to a high degree of uncertainty and risk.

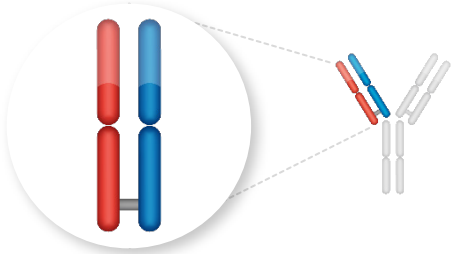
Agenda

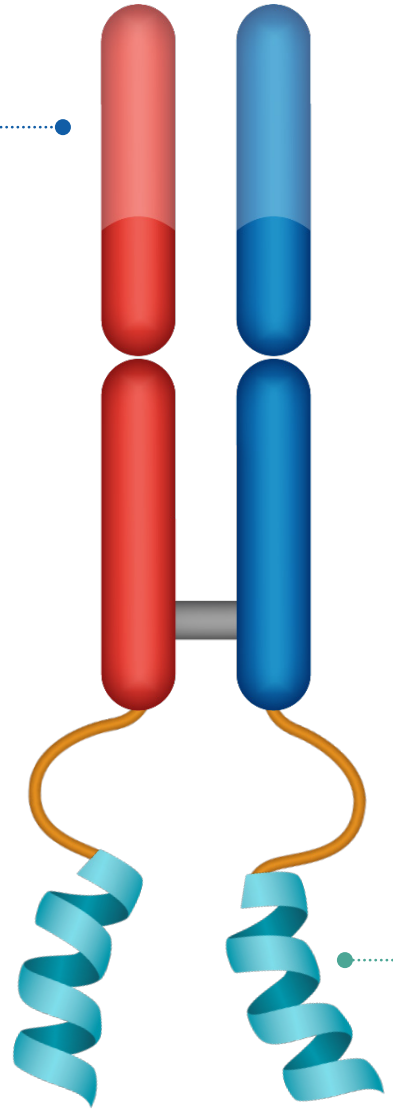
- FORCE platform to enable muscle-targeted delivery of oligonucleotides
- Application of FORCE platform for the treatment of DMD
  - Functional benefit in the *mdx* mouse model
  - FORCE platform achieves high level dystrophin expression across cardiac and skeletal muscle in *mdx* model
- On path to clinical translation with DYNE-251
  - Exon skipping in DMD patient cells
  - Significant exon skipping in NHP cardiac and skeletal muscles
  - NHP toxicology results support advancement of DYNE-251 into the clinic

Dyne FORCE™ Platform: Modern Oligo Therapeutics for Muscle Diseases

ANTIBODY

Proprietary Fab targets TfR1 to enable muscle delivery





LINKER

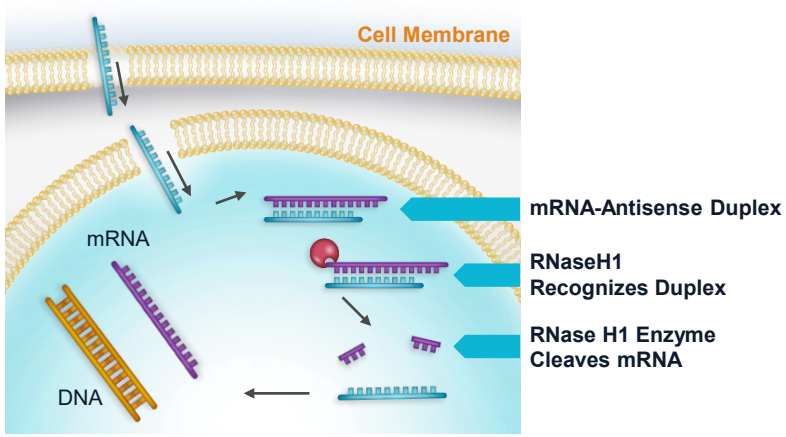
Clinically validated, enables precise conjugation of payloads to a single Fab

PAYLOAD

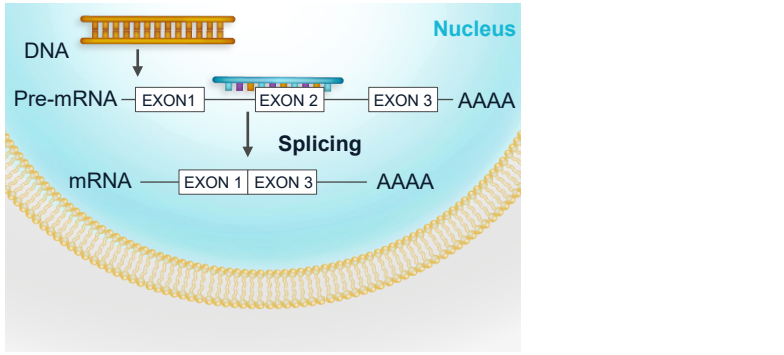
Modularity enables rational selection of payload to target the genetic basis of disease

Rationally Select Payload to Target Genetic Basis of Disease

ASO acts in the nucleus and cytoplasm

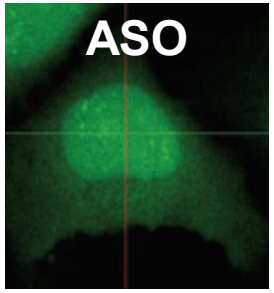


Splice-modulating ASO

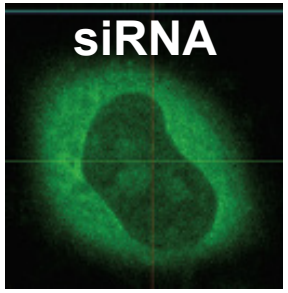


Single-Stranded Antisense

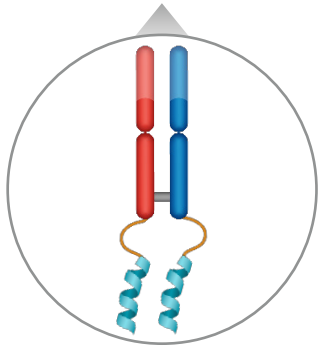
Subcellular distribution of ASO and siRNA



Nuclear localization

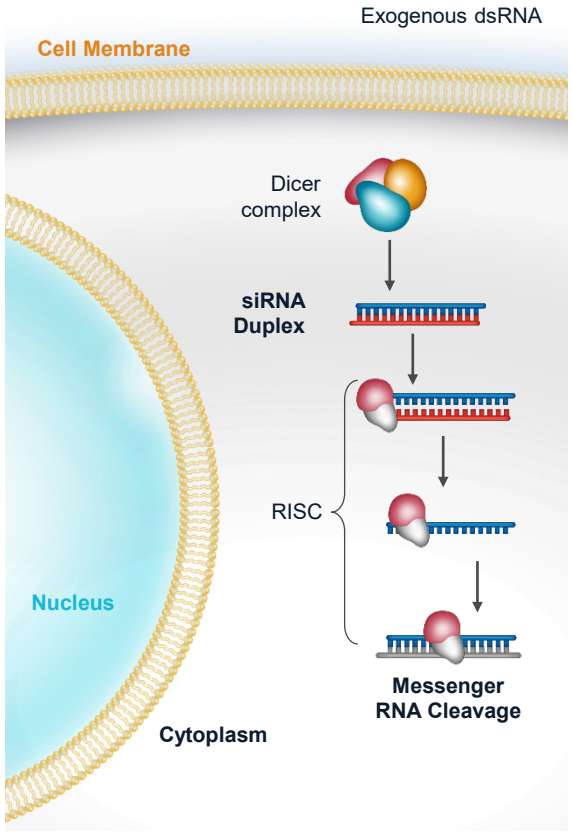


Cytoplasmic localization



FORCE delivers **ASO** payload for nuclear targets, **siRNA** payload for cytoplasmic targets

siRNA acts in the cytoplasm



Double-Stranded Antisense (siRNA)

Adapted from Ohrt T., et al. *Nucleic Acids Res* 2006;34:1369.

FORCE Platform Designed to Deliver Significant Advantages

Targeted Muscle Delivery

Leverages TfR1 expression
on skeletal, cardiac and smooth muscle

Targets Genetic Basis of Disease

Rationally select payloads
to match target biology

Redosable Administration

Potential for individualized patient
titration and longer-term efficacy

Enhanced Tolerability

Targeted delivery limits systemic
drug exposure

Extended Durability

Potential for prolonged disease-modifying
effects, enabling less frequent dosing

Reduced Development and Manufacturing Costs

A single Fab and linker
utilized across all programs

Building a Global DMD Franchise of Transformative Therapies

Overview

- Mutation in the *DMD* gene that encodes for dystrophin
- Onset in first few years of life
- Life expectancy ~30 years

Clinical Presentation

- Muscle weakness
- Progressive loss of function
- Loss of ambulation
- Respiratory/cardiac failure

Population

- ~12,000 - 15,000 (US)
- ~ 25,000 (Europe)

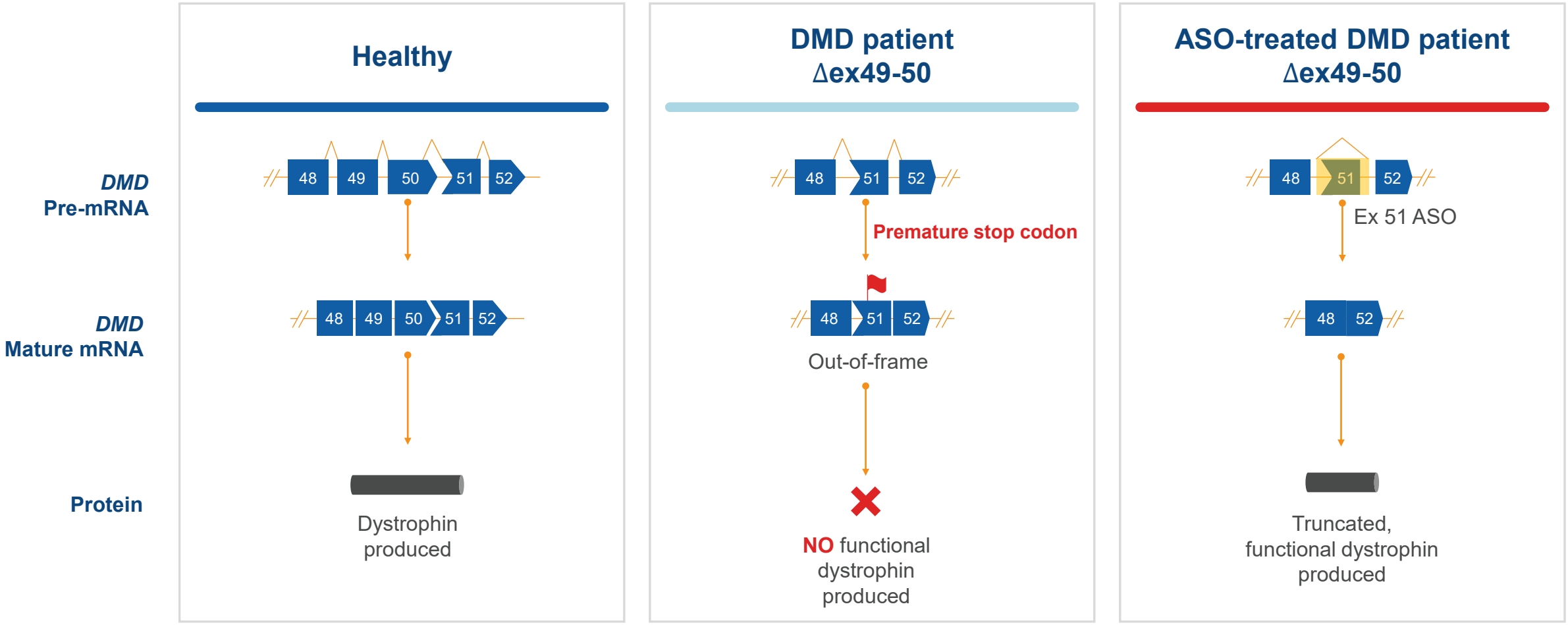
Current Approved Exon 51 Therapies Only Increased Dystrophin Production <1%

OUR APPROACH

Best-in-class Targeted Exon Skipping

Increase dystrophin expression and enable less frequent dosing to potentially **stop or reverse disease progression**

ASO-Mediated Exon Skipping: Mechanism for Disease Correction



Adapted from Lim KRQ., et al. *Drug Des Devel Ther* 2017;11:533.

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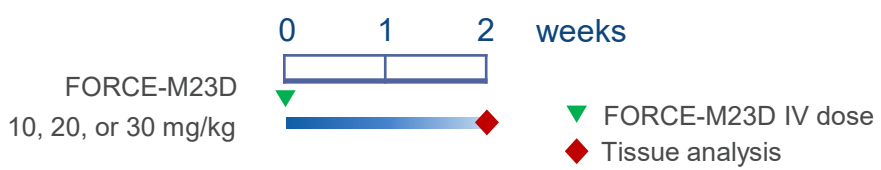
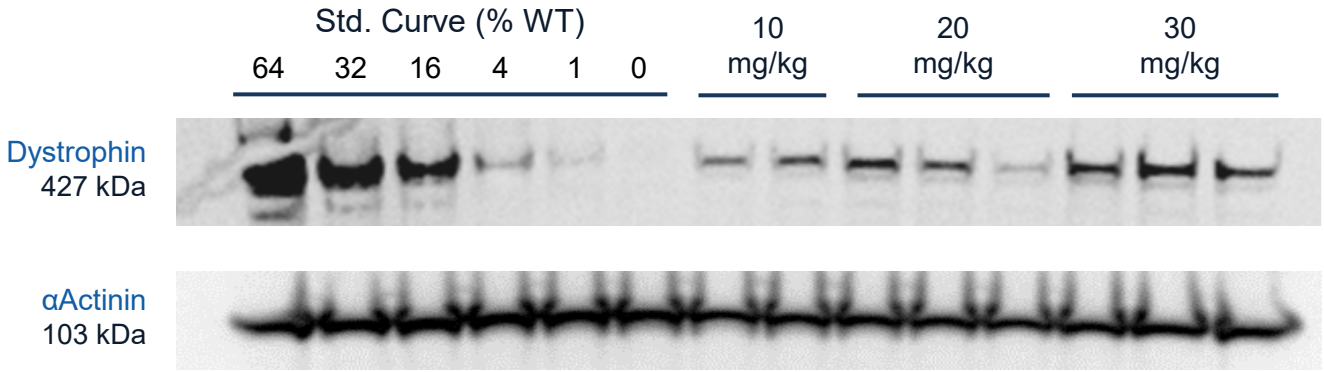
FORCE Dose-Dependently Increased Dystrophin Expression



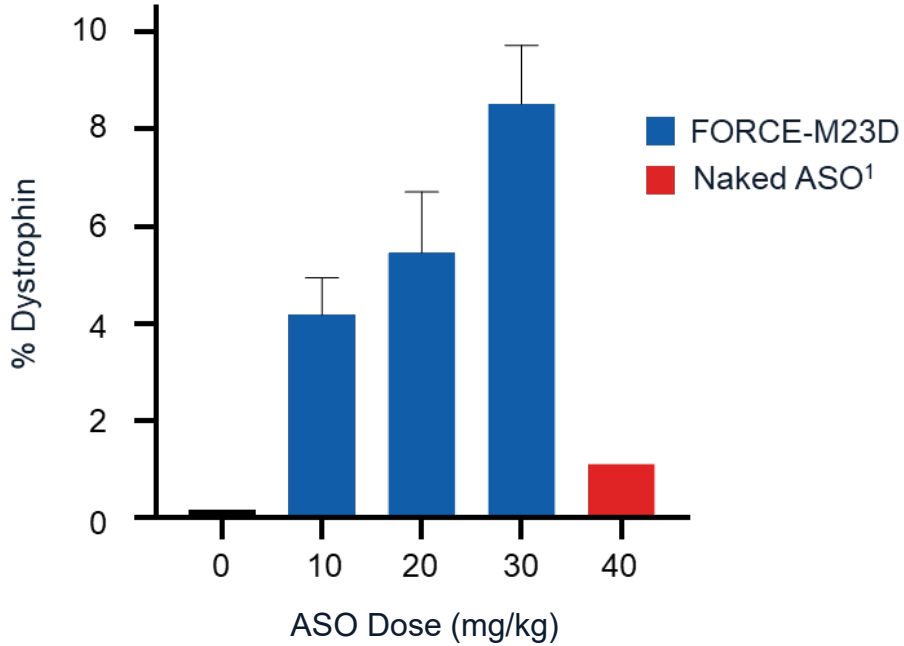
Quadriceps

Dose-Dependent Increase in Dystrophin Expression



Standard curve - Used pooled WT protein and pooled mdx protein,
% indicates amt. of WT spiked into sample

Restored Dystrophin Expression

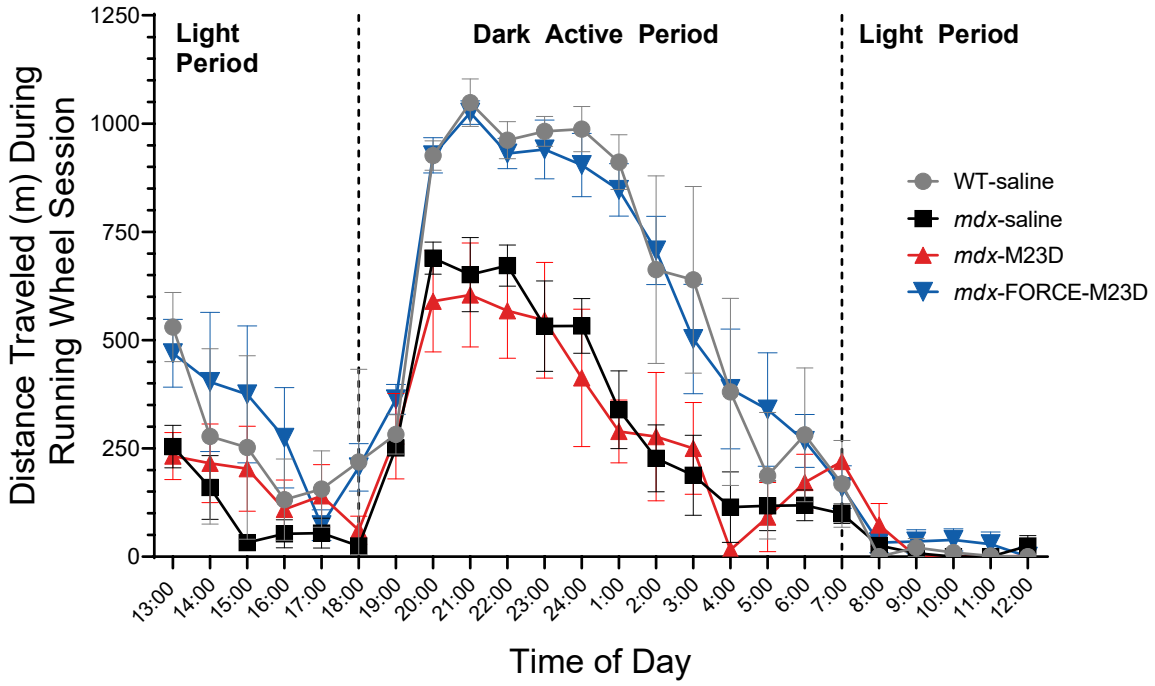


Note: Single IV dose of FORCE-M23D in *mdx* mouse model on day 0; assessment on day 14.
[1]Naked ASO data consist of Sarepta Therapeutics data disclosed in a patent application filed Dec. 13, 2017 after single IV dose.
Subramanian R, et al. *American Society for Cell and Gene Therapy 2020 Annual Meeting.* Abstract 1074.

FORCE Demonstrated Functional Benefit with a Single Dose

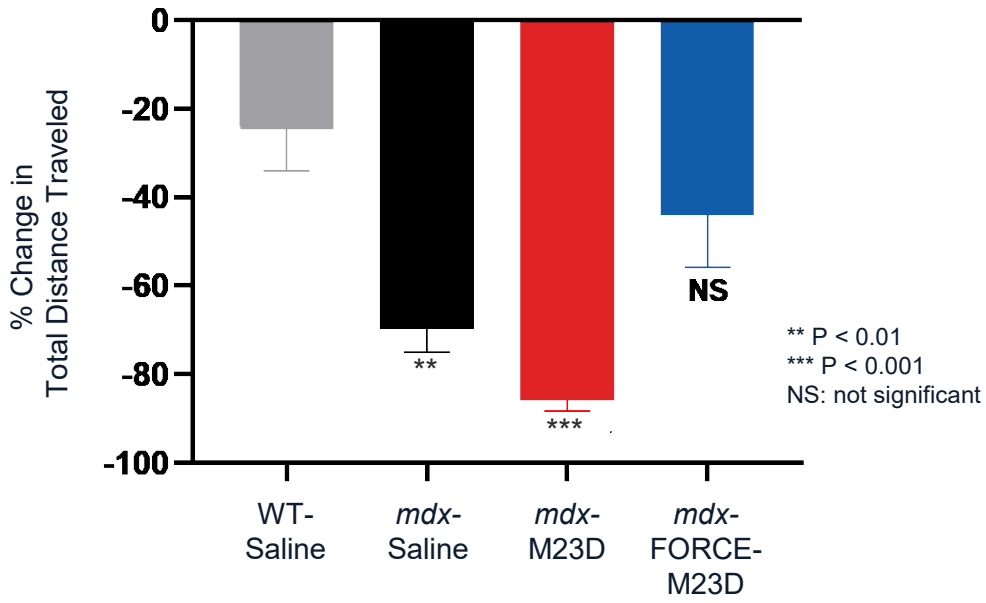
Distance Traveled in Home Cage Running Wheel

(Assessed 4 weeks after treatment)



Distance Traveled in Open Field Following Hind Limb Fatigue Challenge

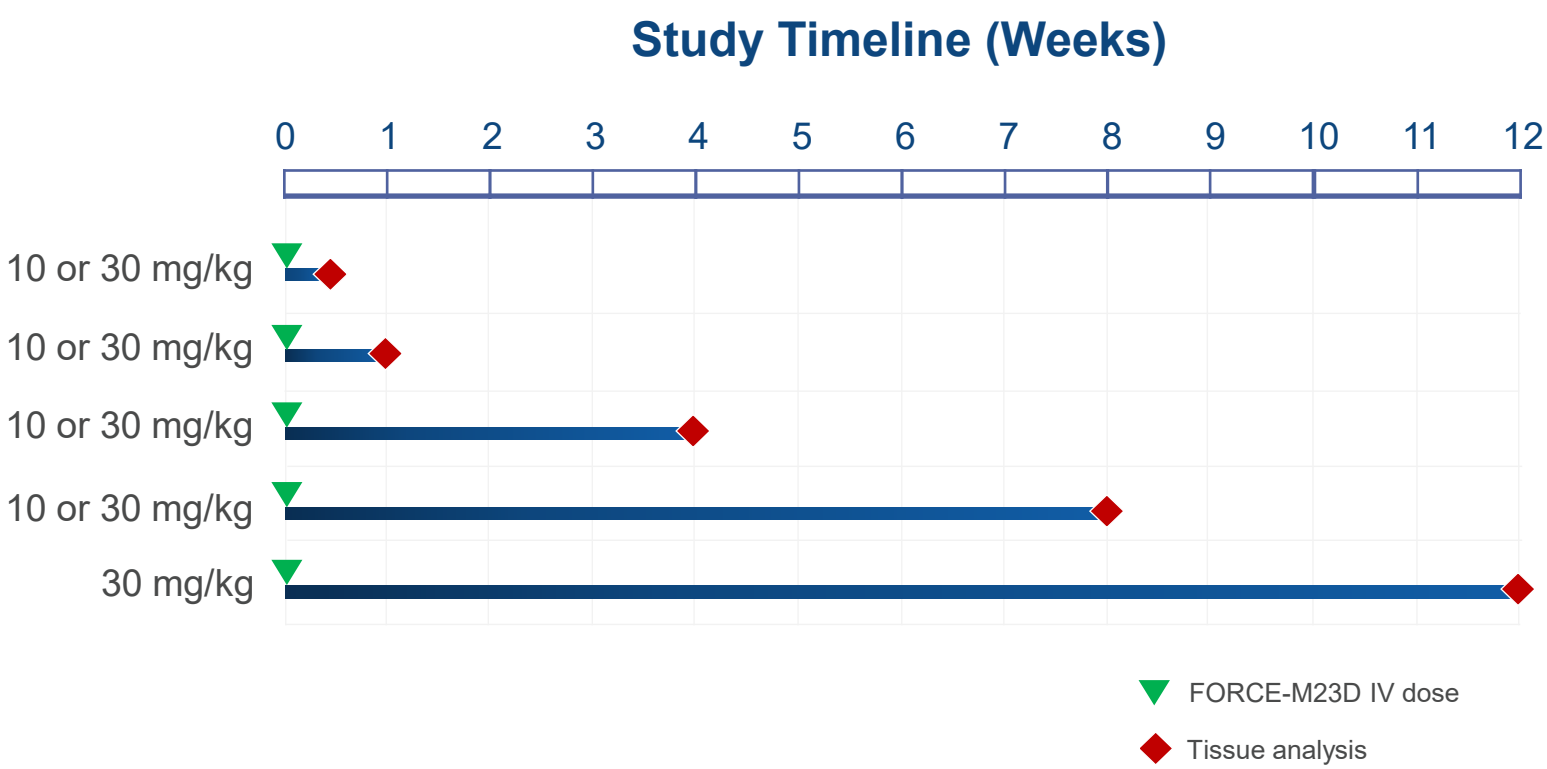
(Assessed 2 weeks after treatment)



Note: Single IV 30 mg/kg dose of FORCE-M23D in *mdx* mouse model. Hind limb fatigue challenge test statistical analysis comparison to wild type (WT) group using one-way ANOVA followed by post-hoc Dunnett's test.

Subramanian R, et al. *American Society for Cell and Gene Therapy 2020 Annual Meeting.* Abstract 1074.

Study Evaluated Dynamic of FORCE on Dystrophin Expression up to 12 Weeks After a Single Dose



Endpoints

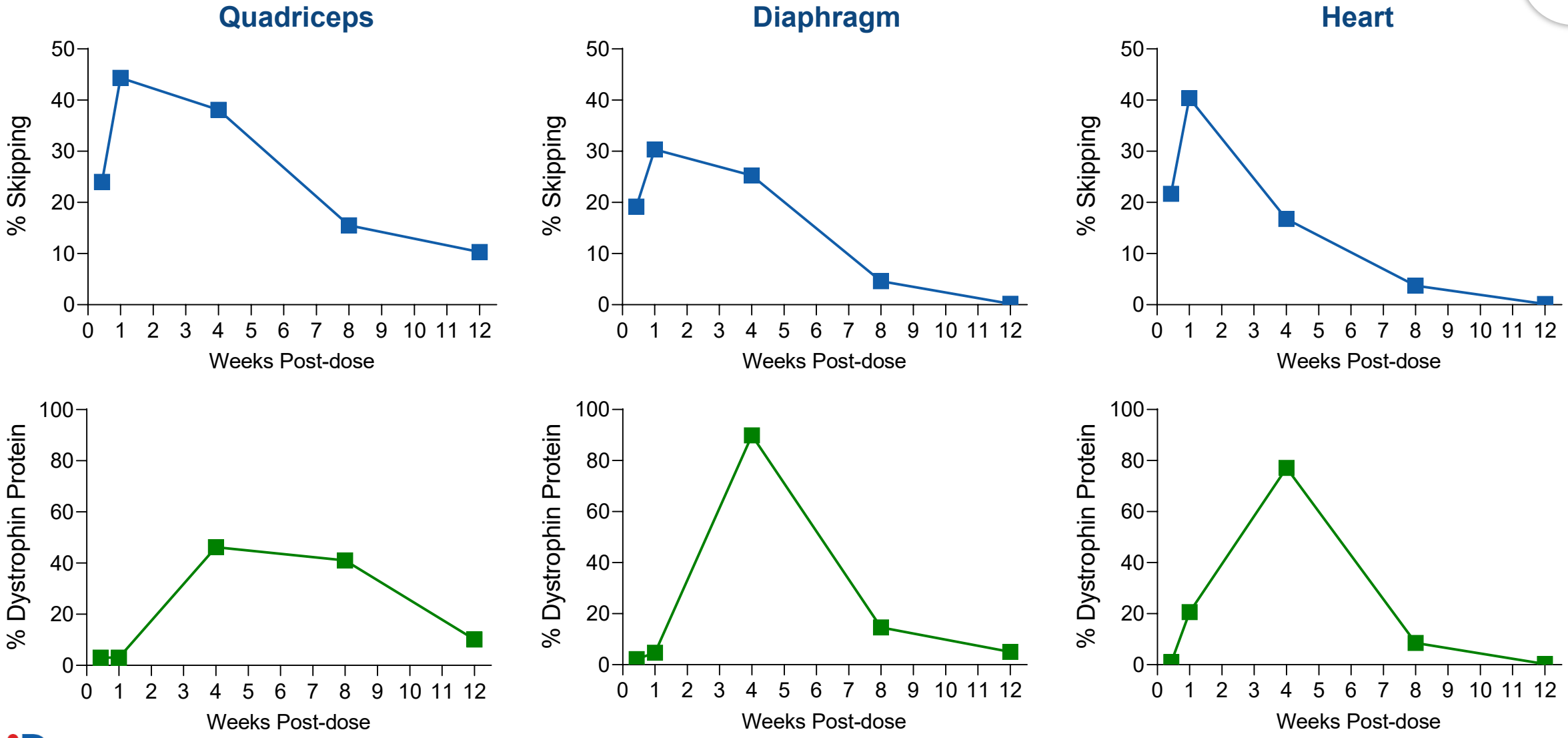
- ASO muscle concentration
- Exon skipping by PCR
- Dystrophin protein by WB
- Dystrophin localization by IF

Tissues analyzed

- Quadriceps
- Diaphragm
- Heart

PCR, polymerase chain reaction; WB, western blot; IF, immunofluorescence

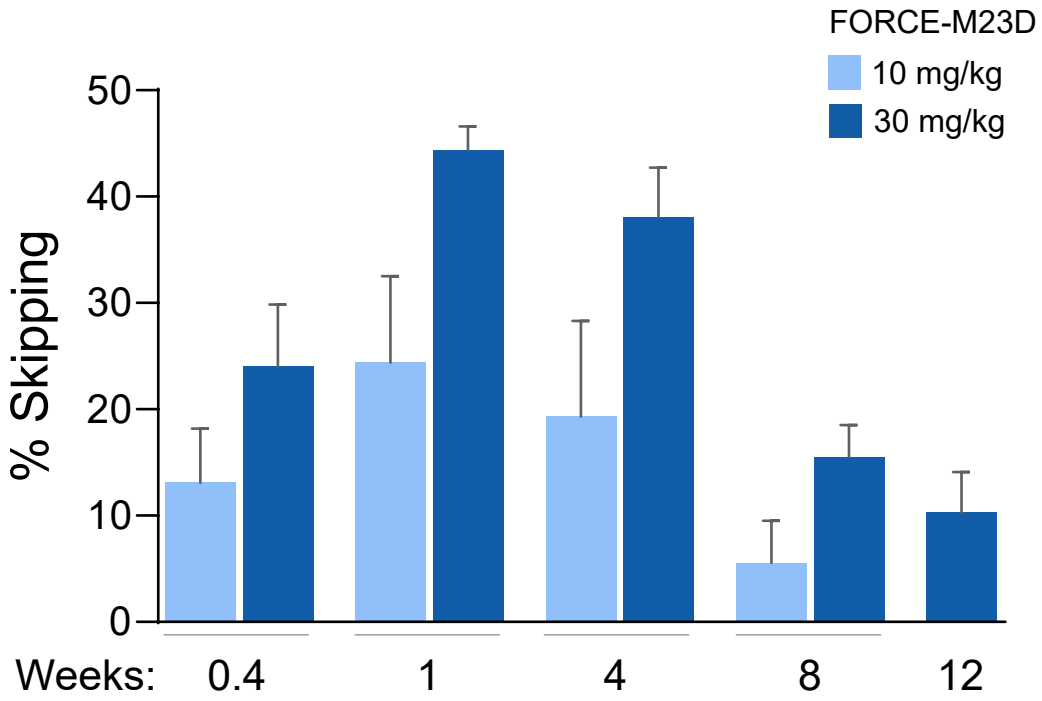
FORCE Achieved Robust and Durable Skipping and Dystrophin Expression in Cardiac and Skeletal Muscle



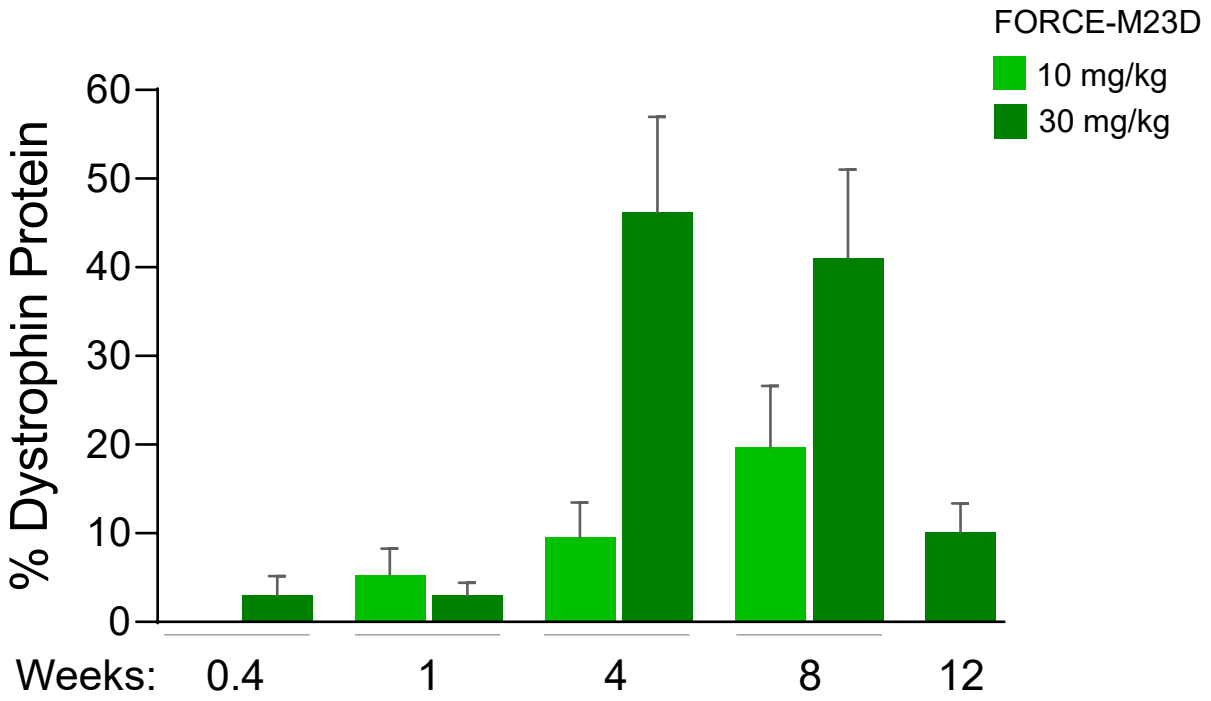
Note: Single IV 30 mg/kg dose of FORCE-M23D in *mdx* mouse model on day 0; N = 3 - 5 per cohort.

FORCE Achieved Robust and Durable Skipping and Expression of Dystrophin in Quadriceps

% Skipping by PCR



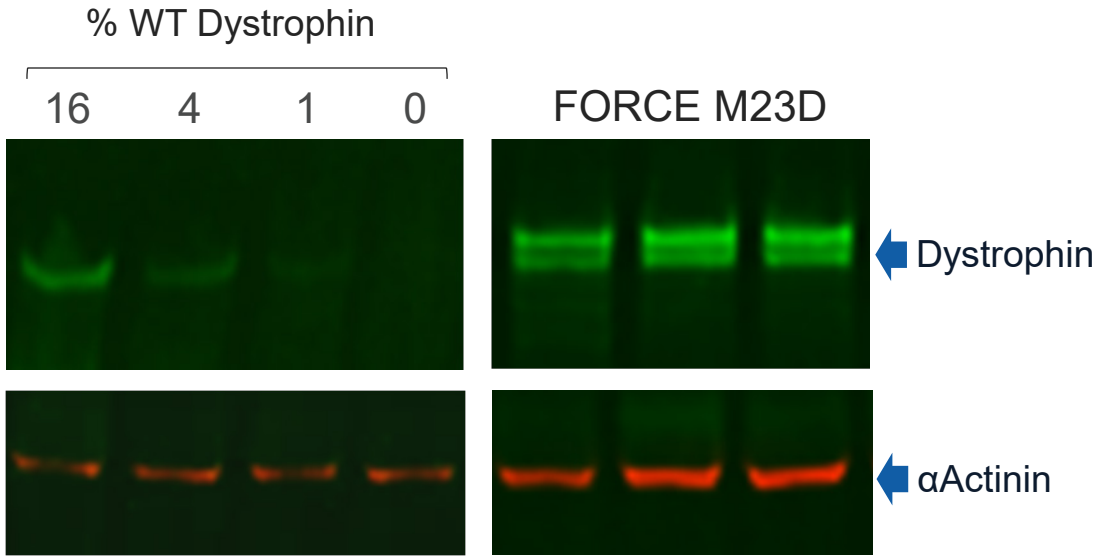
Dystrophin by WB



Note: Single IV dose of FORCE-M23D in *mdx* mouse model on day 0; Data are mean ±SD; N = 4 - 5 per cohort.

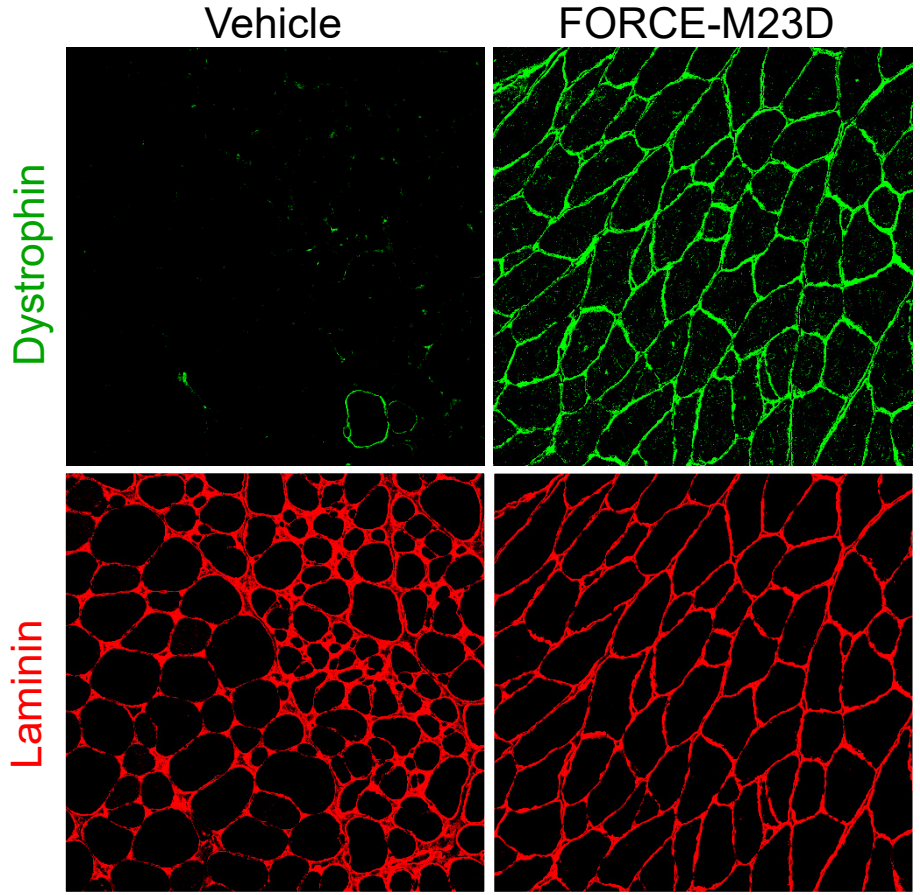
FORCE Achieved Robust Dystrophin Expression and Localization to Sarcolemma in Quadriceps at 8 Weeks

Dystrophin Expression by WB
30 mg/kg 8 Weeks Post-Dose



41% of wild-type dystrophin

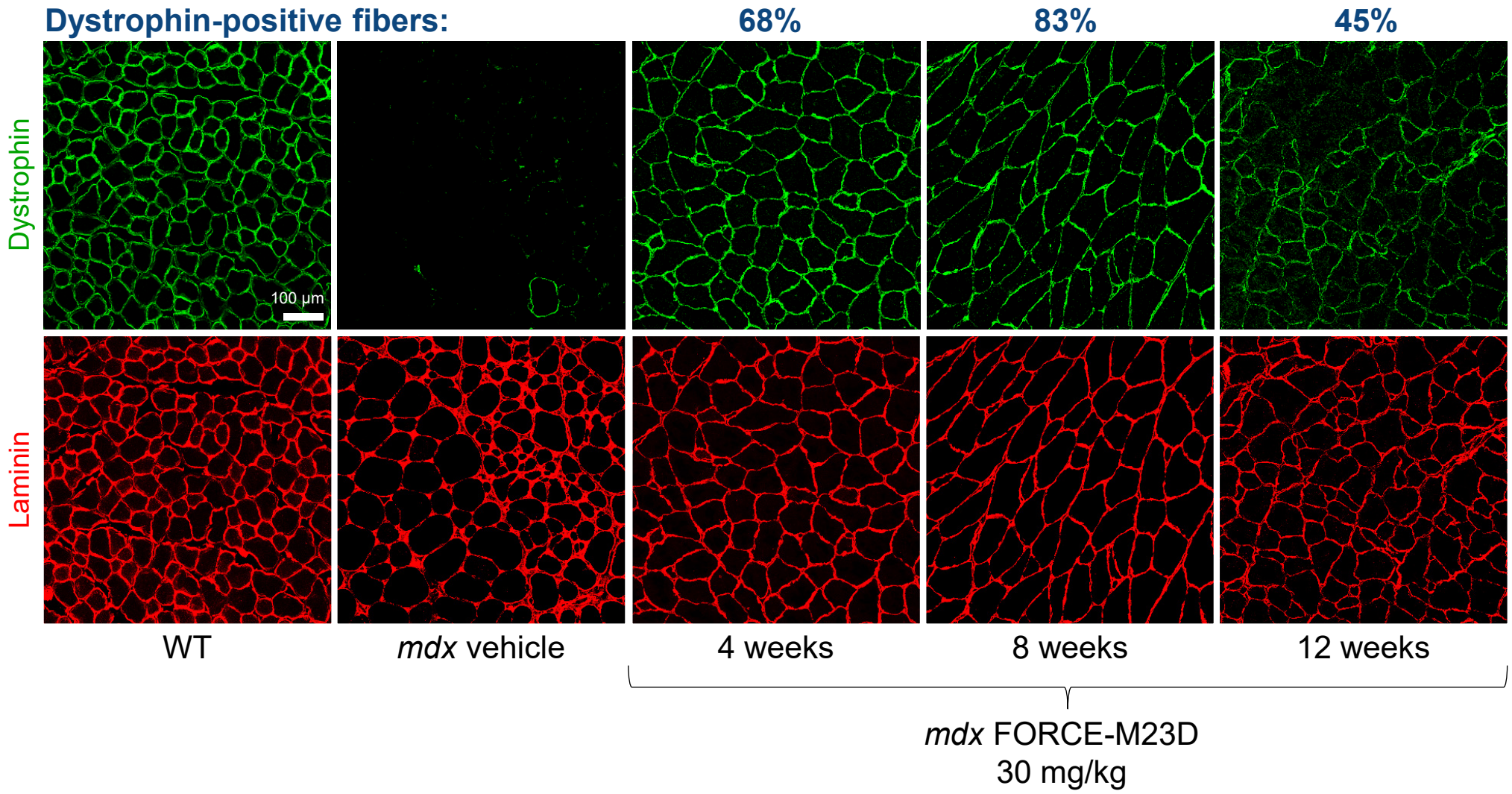
Dystrophin Localization to Sarcolemma



83% dystrophin-positive fibers

Note: Single IV 30 mg/kg dose of FORCE-M23D in *mdx* mouse model on day 0, analysis on week 8.

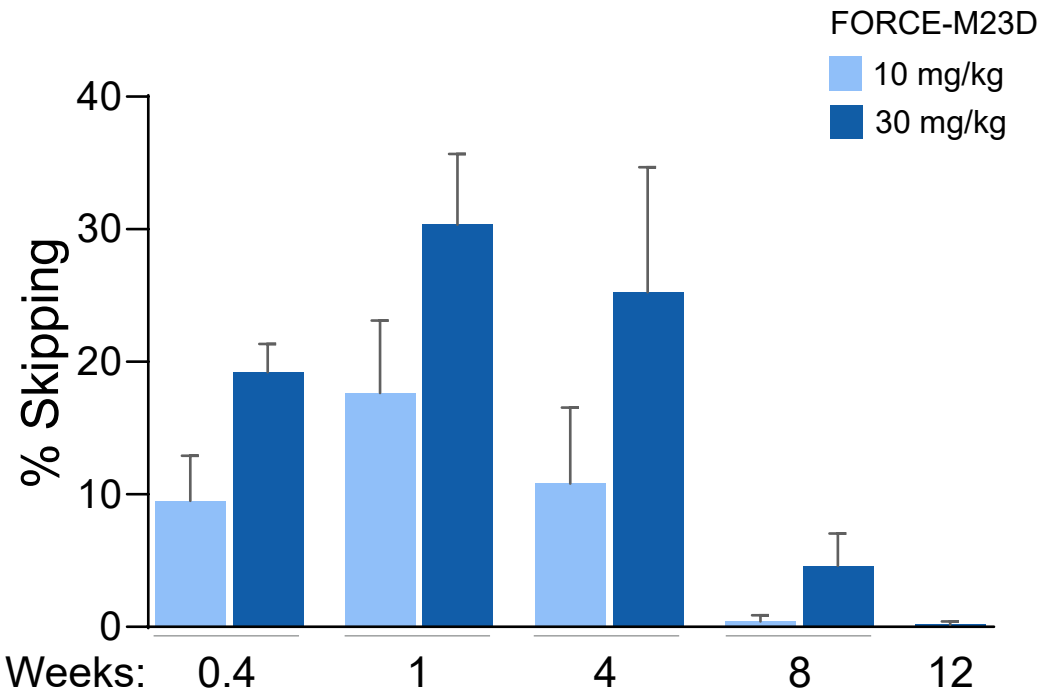
FORCE Achieved Durable Dystrophin Localization to Sarcolemma in Quadriceps



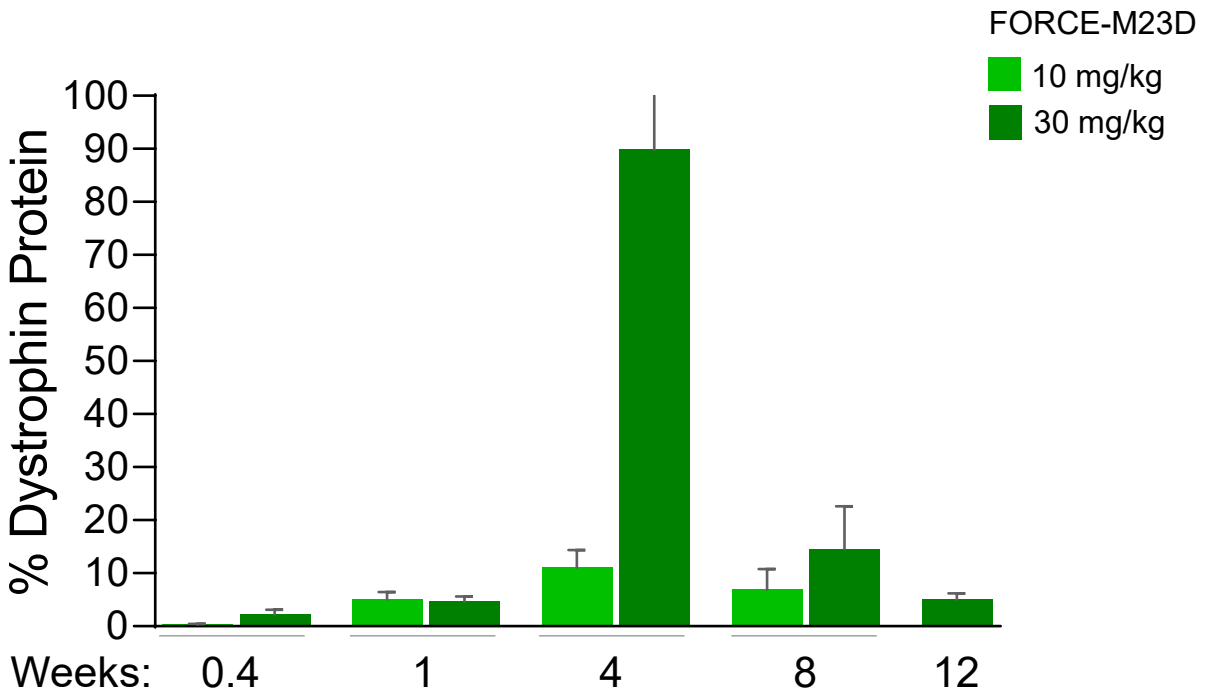
Note: Single IV 30 mg/kg dose of FORCE-M23D in *mdx* mouse model on day 0, analysis at specified time intervals.

FORCE Achieved Robust and Durable Skipping and Expression of Dystrophin in Diaphragm

% Skipping by PCR



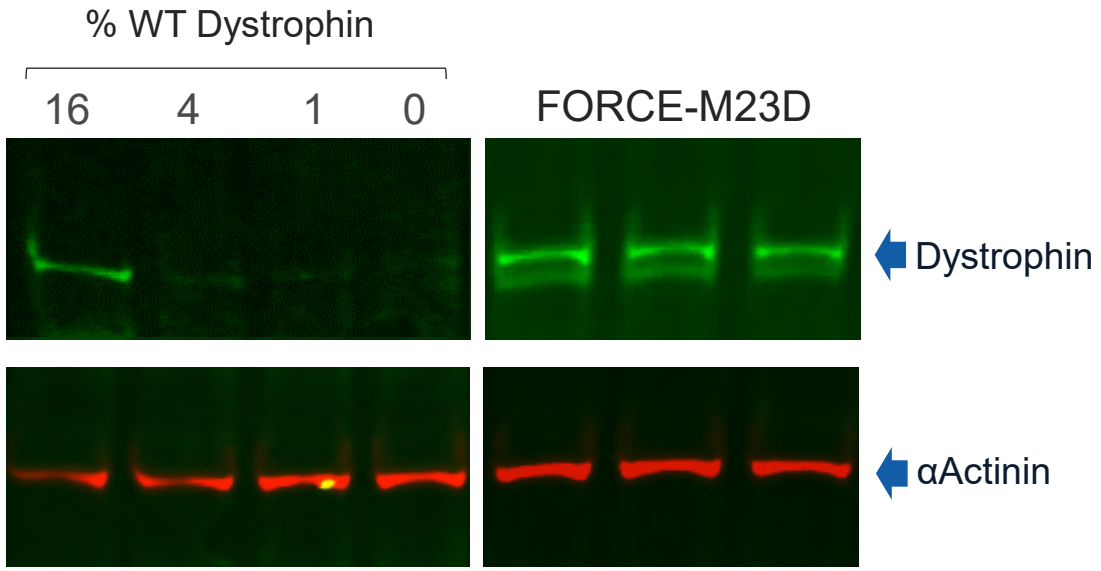
Dystrophin by WB



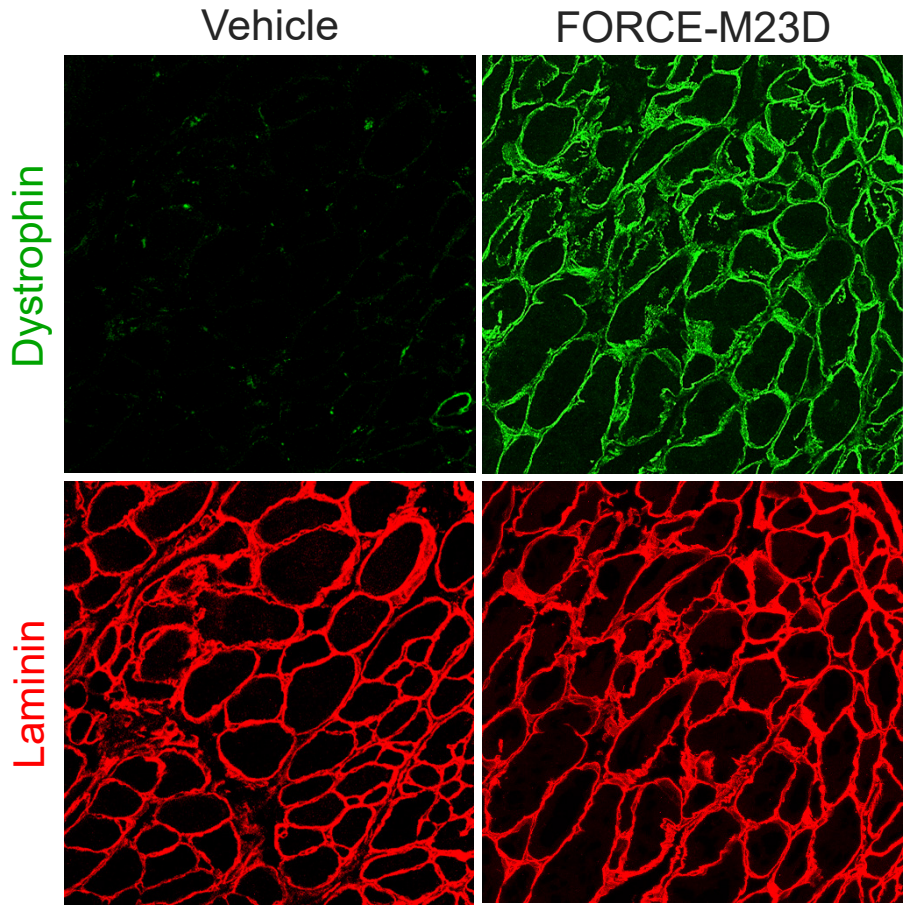
Note: Single IV dose of FORCE-M23D in *mdx* mouse model on day 0. Data are mean ±SD; N = 3 - 5 per cohort.

FORCE Achieved Robust Dystrophin Expression and Localization to Sarcolemma in Diaphragm at 4 Weeks

Dystrophin Expression by WB
30 mg/kg 4 Weeks Post-Dose



Dystrophin Localization to Sarcolemma



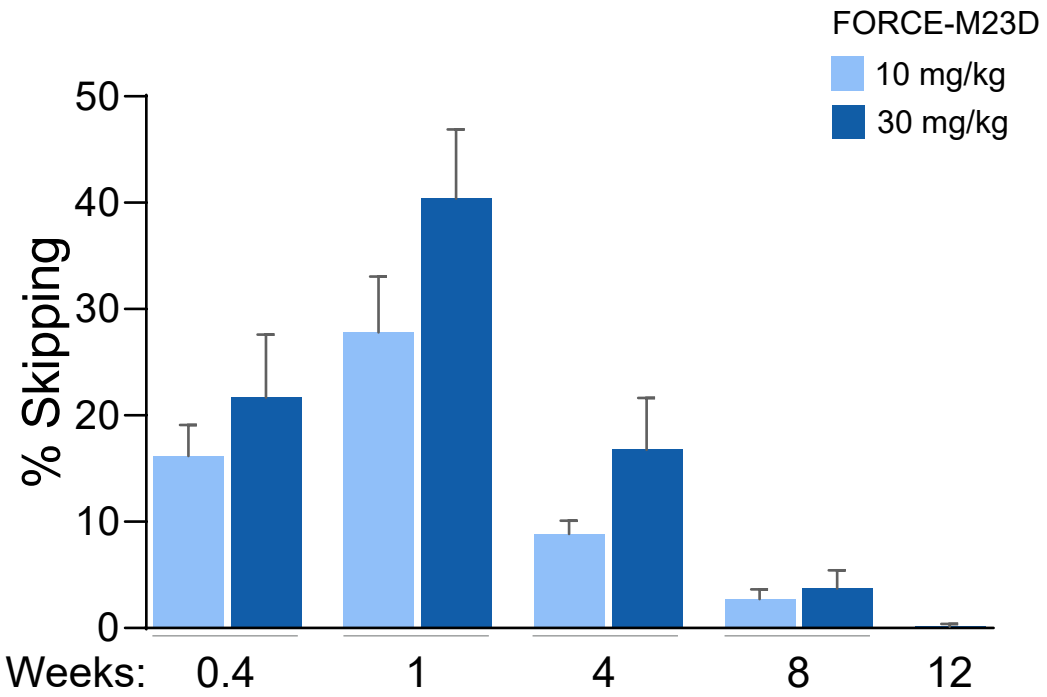
90% of wild-type dystrophin

~80% dystrophin-positive fibers

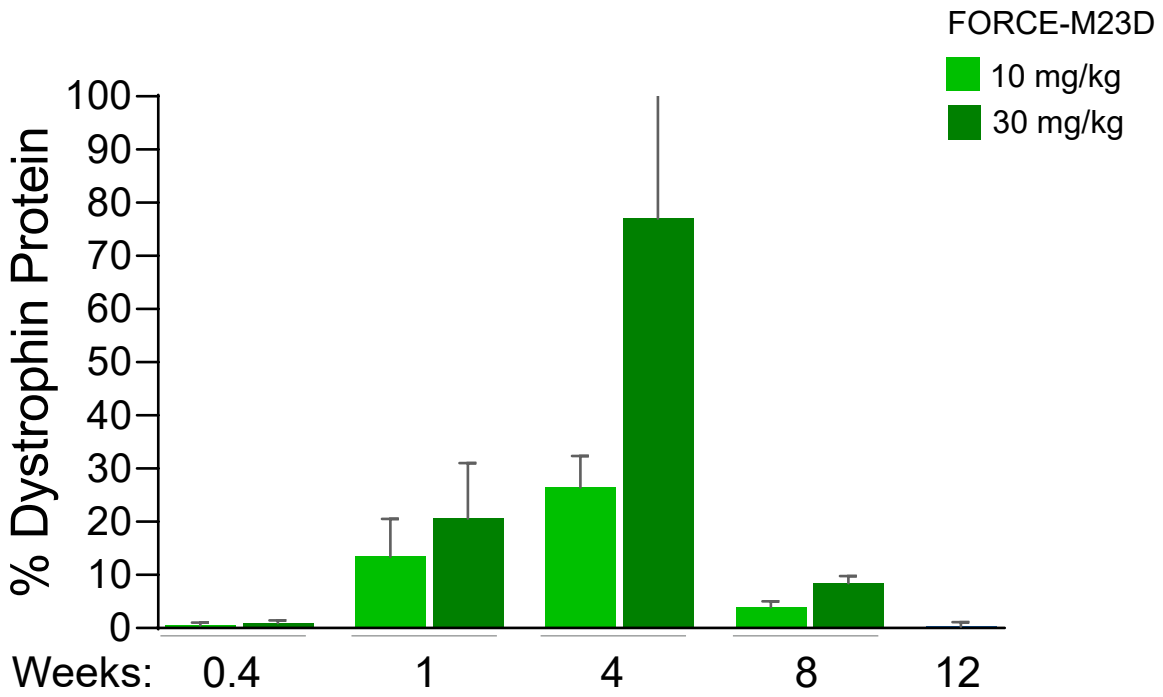
Note: Single IV 30 mg/kg dose of FORCE-M23D in *mdx* mouse model on day 0, analysis on week 4.

FORCE Achieved Robust and Durable Skipping and Expression of Dystrophin in Heart

% Skipping by PCR



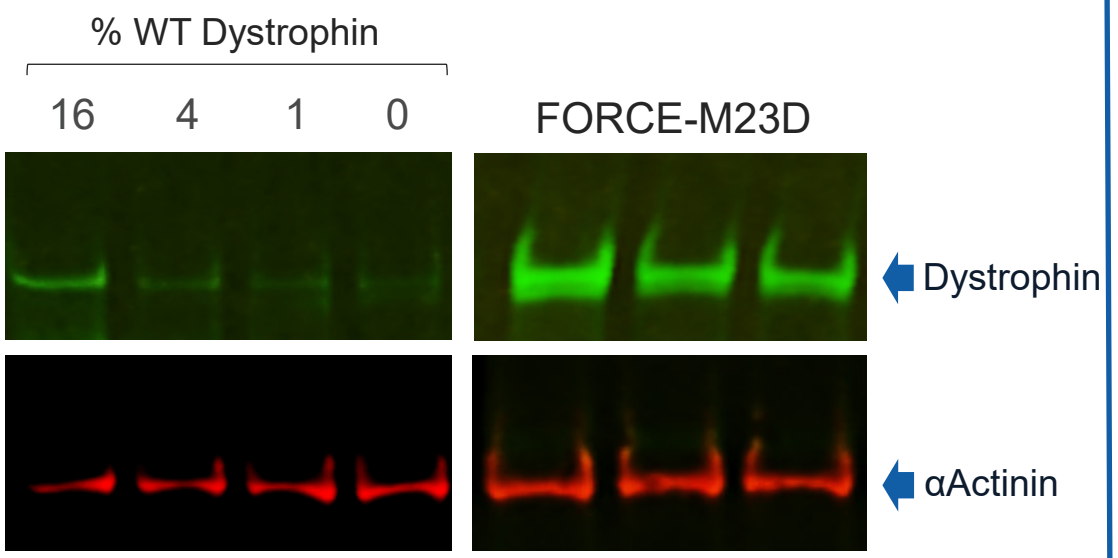
Dystrophin by WB



Note: Single IV dose of FORCE-M23D in *mdx* mouse model on day 0. Data are mean ±SD; N = 4 - 5 per cohort.

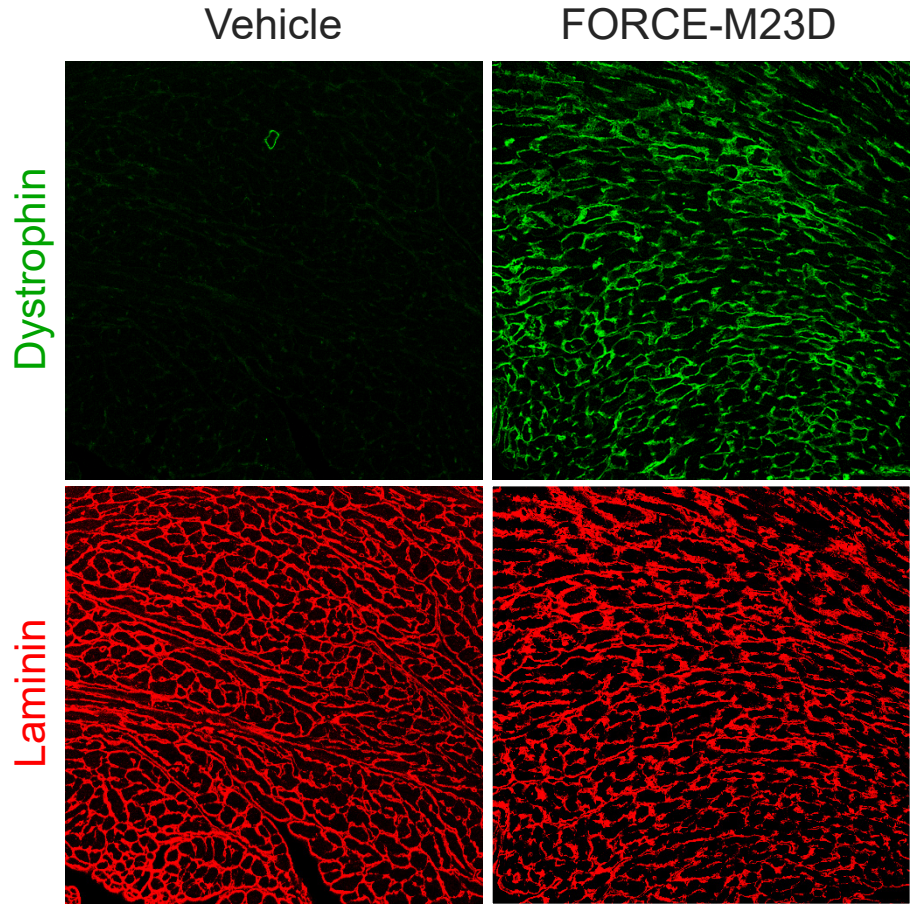
FORCE Achieved Robust Dystrophin Expression and Localization to Sarcolemma in Heart at 4 Weeks

Dystrophin Expression by WB
30 mg/kg 4 Weeks Post-Dose



78% of wild-type dystrophin

Dystrophin Localization to Sarcolemma



~80% dystrophin-positive fibers

Note: Single IV 30 mg/kg dose of FORCE-M23D in *mdx* mouse model on day 0, analysis on week 4.

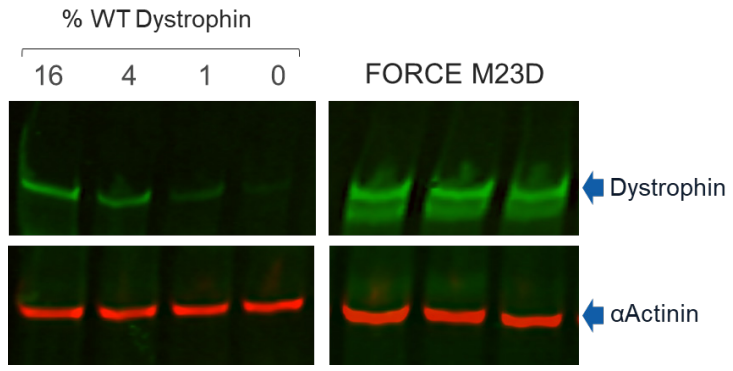
FORCE Achieved Robust and Durable Dystrophin Expression and Sarcolemma Localization in Muscle

Quadriceps

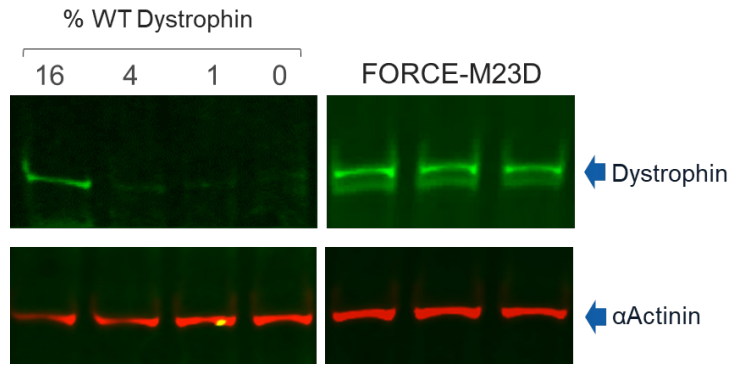
Diaphragm

Heart

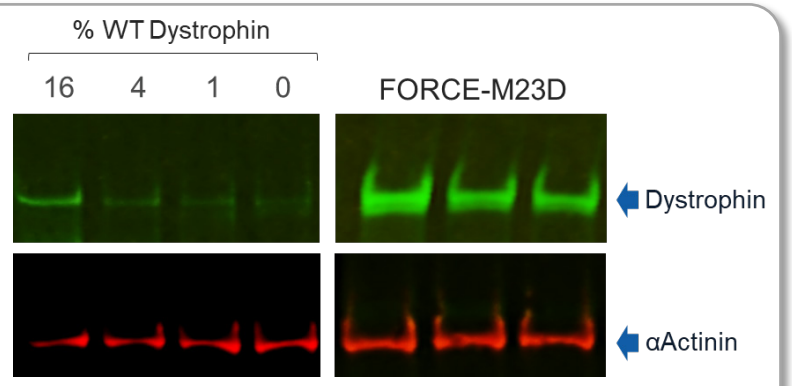
Dystrophin expression by WB



46% of wild-type dystrophin

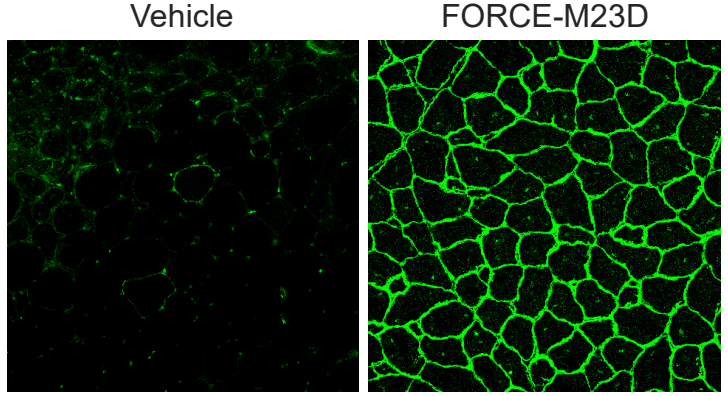


90% of wild-type dystrophin

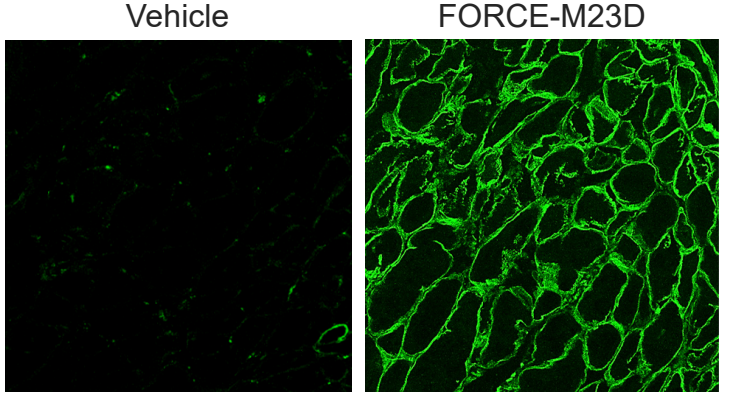


78% of wild-type dystrophin

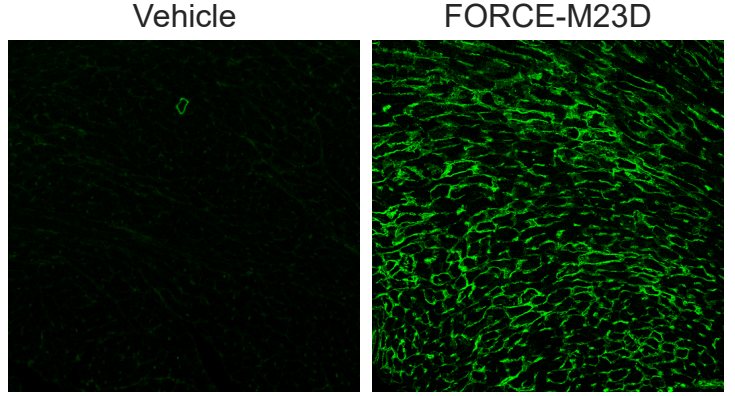
Dystrophin localization to sarcolemma



68% dystrophin-positive fibers



~80% dystrophin-positive fibers



~80% dystrophin-positive fibers

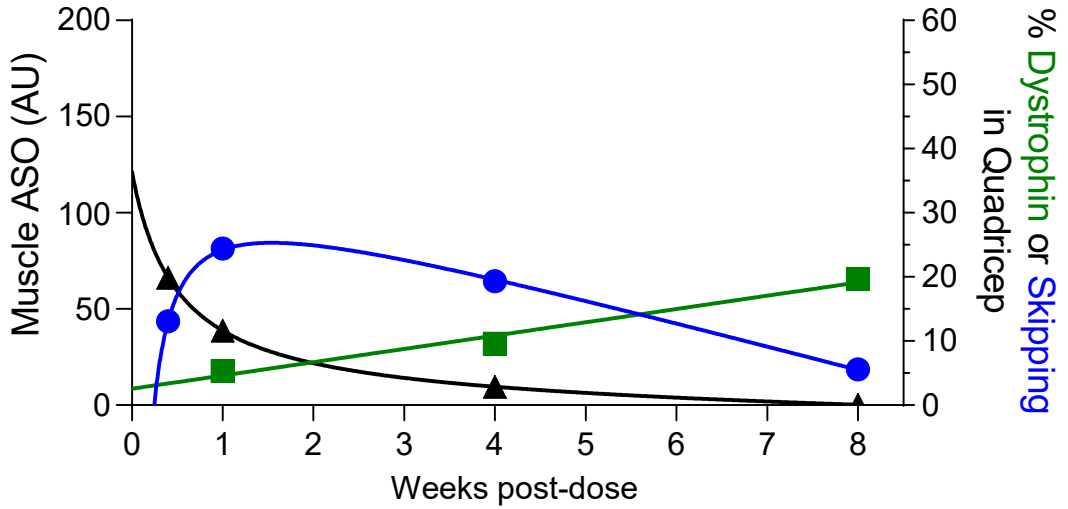
Note: Single IV 30 mg/kg dose of FORCE-M23D in *mdx* mouse model on day 0; analysis on week 4 for all muscles. N= 3 - 5 per cohort.

FORCE Distinctive Pharmacokinetic Profile Delivered Substantial and Durable Dystrophin Expression with a Single Dose

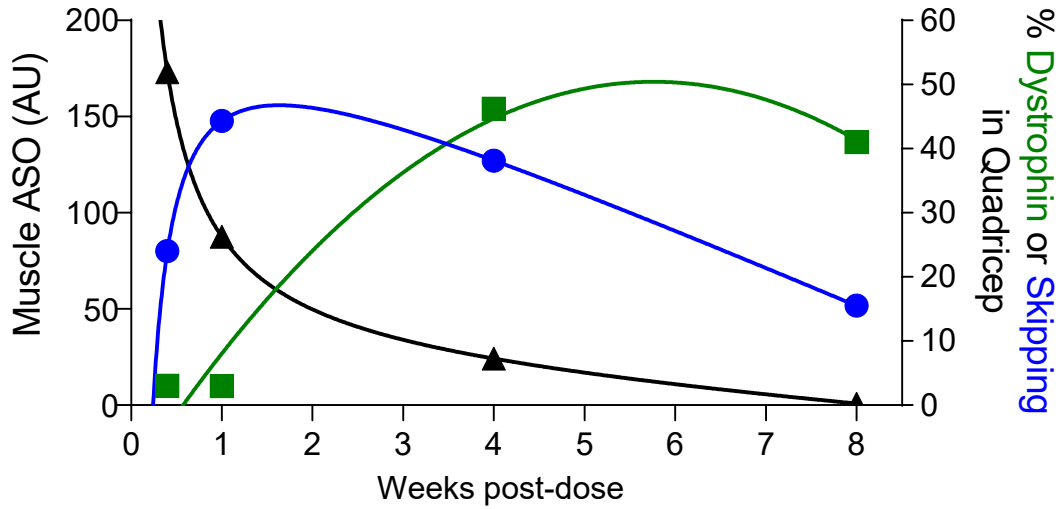
Pharmacokinetics | **Pharmacodynamics**

Muscle ASO concentration → Exon skipping → Dystrophin restoration

FORCE-M23D 10 mg/kg



FORCE-M23D 30 mg/kg



Muscle ASO | Skipping | Dystrophin

Note: Single IV dose of FORCE-M23D in *mdx* mouse model on day 0. N = 3 - 5 per cohort. ASO levels expressed as arbitrary units (AU).

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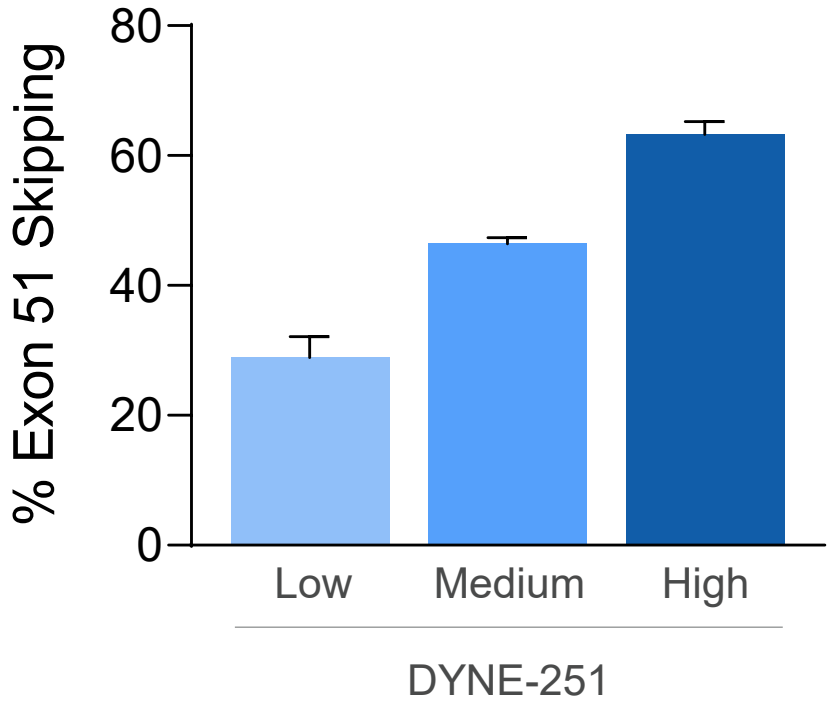
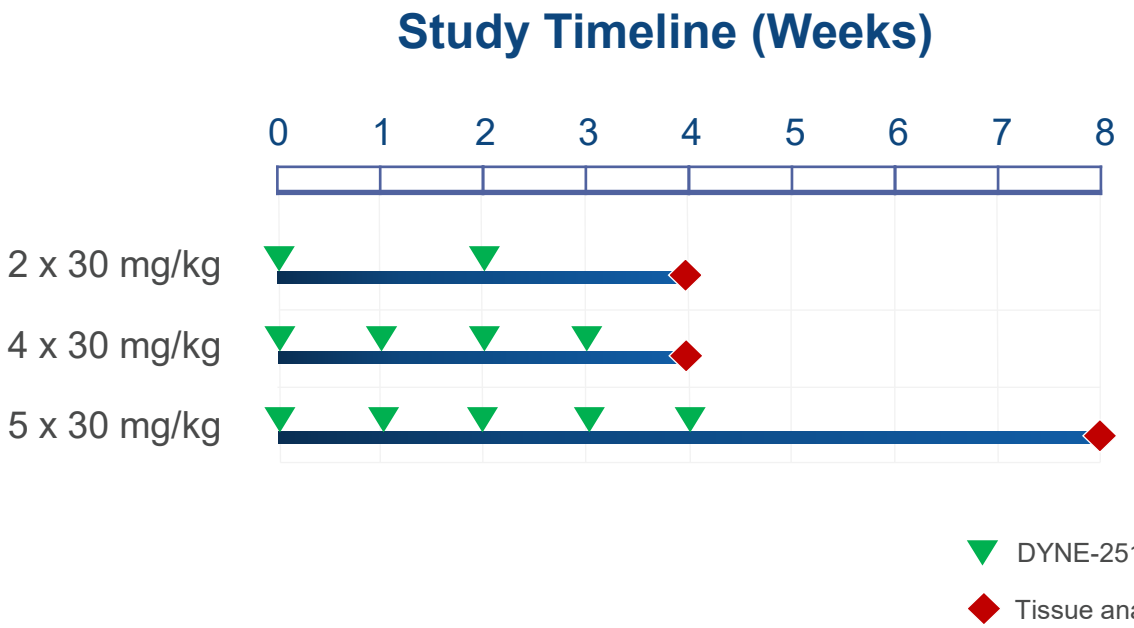
  - NHP toxicology results support advancement of DYNE-251 into the clinic

DYNE-251 Achieved Robust and Dose-Dependent Exon 51 Skipping in DMD Patient Myotubes



Note: DYNE-251 exposure in DMD patient myotubes for 10 days. N = 3.

Dose Regimen Study in NHPs to Inform Clinical Dose



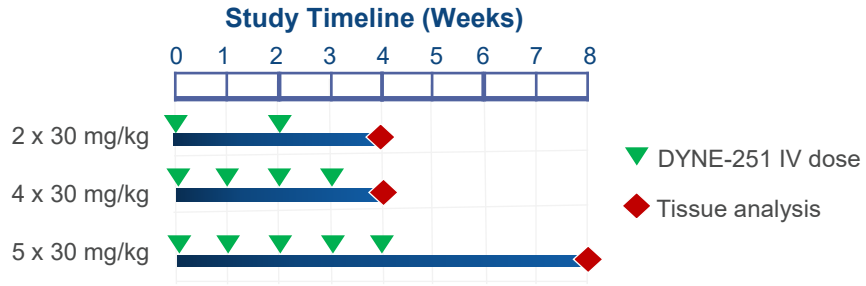
Endpoints

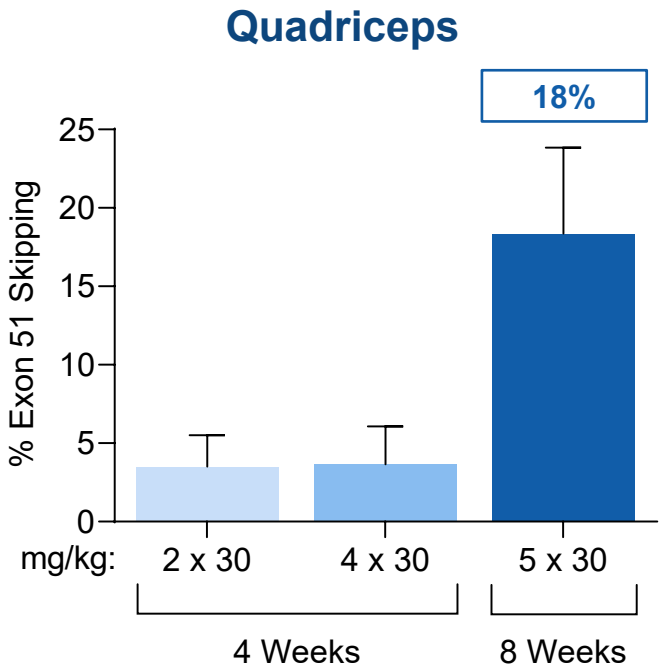
- Exon skipping by PCR

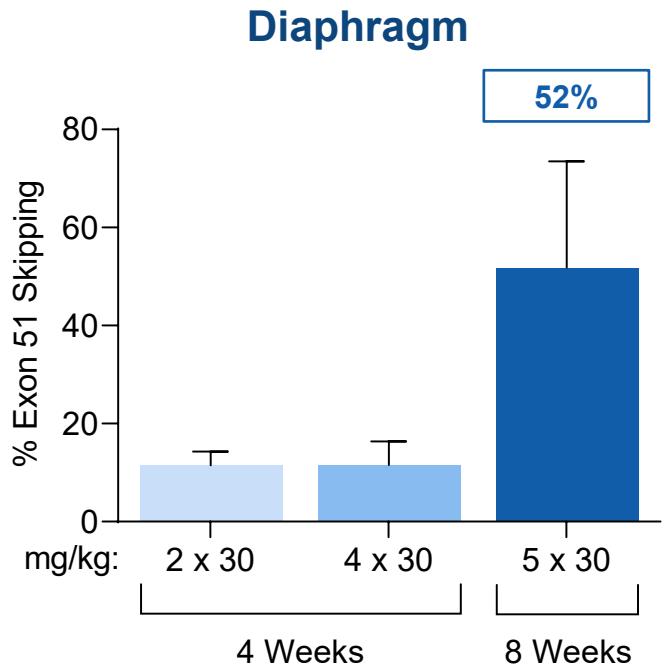
Tissues analyzed

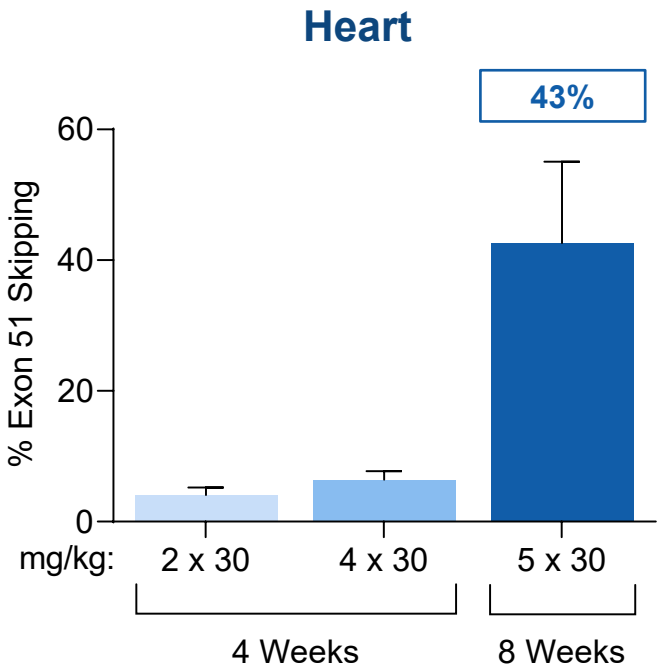
- Quadriceps
- Diaphragm
- Heart

DYNE-251 Achieved Robust Exon Skipping in NHP Skeletal and Cardiac Muscles









Note: Data are means ± SD. N = 3 - 5 per cohort.

DYNE-251 NHP GLP Toxicology Results Demonstrated Favorable Safety Profile That Support Advancement to Clinic

- No dose limiting toxicity observed after five weekly doses up to a maximally feasible dose
- No changes in cardiac, respiratory, neurologic or ophthalmic endpoints
- No effect on kidney function
- No effect on liver function
- No effect on coagulation
- NOAEL was identified at the highest dose tested

NOAEL: No-observed-adverse-effect level

Summary

- In *mdx* mouse model FORCE achieved:
  - Dose-dependent, robust, and durable skipping and dystrophin expression
  - Long-lasting dystrophin localization to sarcolemma
  - Functional benefit
- DYNE-251 demonstrated robust and dose-dependent exon 51 skipping in DMD patient myotubes
- DYNE-251 demonstrated robust exon skipping in NHP cardiac and skeletal muscles
- NHP toxicology results support advancement of DYNE-251 into the clinic

Three INDs planned between Q4 2021 - Q4 2022

Targeting the genetic basis of serious muscle diseases to

STOP OR REVERSE DISEASE PROGRESSION

FORCE PLATFORM	**Robust** PIPELINE	**Delivering** FOR PATIENTS	**Exceptional** TEAM